



Watch engineers play Tetris

Watch engineers from North Carolina State University play Tetris using the breathable electronic wearable. <http://digit.in/Sep20-93>



Harvesting mechanical energy

Read more about the fabric that uses sun and wind to potentially power devices. <http://digit.in/Sep20-94>

Credit: Cornell University



The FingerTrak device

oped a device called FingerTrak, a wrist mounted device that can find out the positions, configurations and orientations of all the five fingers. This technology can allow sign language users to communicate with computers using hand gestures. There are four tiny cameras on the wristband, that captures silhouettes of the fingers, which are then processed using a deep neural network, which then builds a three dimensional hand. The technology is able to continuously track the fingers even when the user is interacting with other objects. FingerTrak is also sensitive enough for biomedical applications, and can be used to help patients with Parkinson's and Alzheimer's.

DOES IT BREATHE?

Wearable electronics have been made stretchable, bendable and flexible, but these are still uncomfortable to wear because they are films covering the skin that are not breathable. Researchers from North Carolina State University have recently developed a process that allows for production of breathable electronics at scale. The approach creates a stretchable

polymer film, with holes spaced evenly across the surface, that allows sweat from the skin to evaporate. The film is then dipped in a solution with silver nanowires contained in them to give the "smart" coating, after which they are treated with heat to seal the nanowires in place. The resulting film is permeable to water vapour, conductive to electricity, has optical transmission properties, and is only a few micrometres thick. To test the efficacy of the material, the researchers created two prototype devices, one each for the intended use cases, for biomedical devices and human machine interfaces.

The first prototype was embedded with electrodes that could be used for measuring electrocardiography (ECG) and electromyography (EMG). These are comfortable to wear for long periods of time and do not cause irritation on the skin. The second is a touch sensing device for HMI, and is a wristband embedded with control buttons. The researchers used it to play Tetris, but there are a number of other potential applications for the technology. The most important thing here is the gas permeability of the material, which allows for people using these devices to comfortably wear them for long periods of time.

One of the problems with wearables right now is

that although they have batteries that last longer than say smartphones, it is still a pain to use them. You are likely to forget that the shoe mounted step tracker needs a battery replacement, or forget to charge the GPS tracker in your bag, or if you have one of those posture correction devices, quit using them after the novelty wears off. Researchers from Georgia Tech have solved this problem by embedding the power generation properties into the fabric itself, which is both stretchable and breathable. The fabric was actually made in a regular loom, using fiber-based triboelectric nanogenerators and solar cells made out of polymers. The triboelectric nanogenerators generate electricity from any kind of motion such as vibrations, rotations or sliding. This



Credit: GATech

The GATech electricity harvesting fabric

means that any kind of movement by the individual wearing the material generates electricity. Even if a breeze is blowing, the material generates electricity. Additionally, the solar cells can extract more electricity from sunlight. This fabric can be interwoven with wool, and used to make clothing that generates power, or curtains and tents. This means that your sweater may be able to power up your next smartphone, and the fabric can also potentially have other electronic devices embedded within them, with plenty of power easily available. Both the textile and the electrodes used in the method are easily available and cheap to produce. The researchers are now working on protecting the fabric from rain and moisture. **A**

Credit: Yong Zhu, Matt Shipman / NCS



The breathable touch sensor